

- 6 (a) A student constructed a table to compare the structural features of a plant cell, a prokaryotic cell and a virus.

Complete Table 6.1.

Table 6.1

feature	plant cell	prokaryotic cell	virus
external structure	cell wall composed of cellulose	cell wall composed of	capsid composed of
size of ribosomes	80S and 70S		no ribosomes
nucleic acids	DNA and RNA	DNA and RNA	

[4]

- (b) The cholera bacterium releases a protein toxin called choleraen. The toxin causes the loss of chloride ions and water from epithelial cells into the lumen of the intestine.

Fig. 6.1 shows the events that occur in cells lining the intestine when choleraen binds to the membrane of one of these cells.

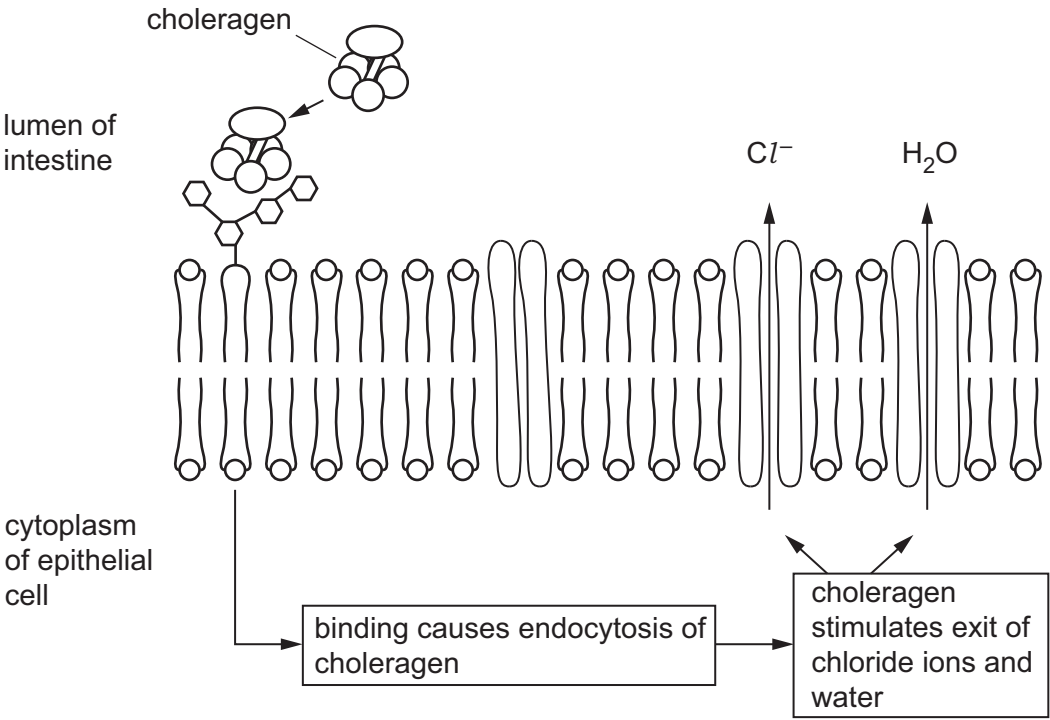


Fig. 6.1



(i) State **one** likely source of an outbreak of cholera.

..... [1]

(ii) With reference to Fig. 6.1, state why cholera toxin molecules are described as having quaternary structure.

..... [1]

(iii) State the type of cell membrane component that forms the receptor for cholera toxin.

..... [1]

(iv) The process by which chloride ions leave the epithelial cell requires energy.

Name the phosphorylated nucleotide that is needed for this process.

..... [1]

(v) Explain why water also moves from epithelial cells into the lumen of the intestine when cholera toxin is present.

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..... [2]

[Total: 10]